Software Requirements Specification

for

Case Management System

Deliverable 1

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Table of Contents

1. Introduction

1.1 Purpose

The purpose of this document is to provide a comprehensive Software Requirements Specification (SRS) for the development of the Legal Case Management System for Pakistan. This document outlines the system's functional and non-functional requirements, design constraints, and other critical aspects that will guide the development process. The goal is to create a robust, user-friendly, and secure platform that streamlines legal case management, reduces administrative burden, and improves the overall efficiency of legal processes in Pakistan.

1.2 Document Conventions

This document follows standard software engineering conventions to ensure clarity and consistency. Key terms and acronyms are defined upon their first use. Requirements are numbered sequentially and categorized into functional and non-functional requirements. Code snippets, if any, are presented in monospaced font. Diagrams are labeled and referenced appropriately in the text. All references to external documents or standards are listed in the References section.

1.3 Intended Audience and Reading Suggestions

This document is intended for all stakeholders involved in the development, deployment, and use of the Legal Case Management System. The primary audience includes software developers, project managers, legal professionals, and system administrators. For developers, it is recommended to focus on Sections 3 (Functional Requirements) and 4 (System Architecture). Legal professionals may find Section 2 (Overall Description) most relevant, as it outlines the system's objectives and key features. Project managers should review the entire document to ensure alignment with project goals and timelines.

1.4 Product Scope

The Legal Case Management System is designed to automate and streamline the management of legal cases in Pakistan. The system will provide functionalities for case tracking, document management, court scheduling, client communication, and reporting. It will integrate with existing legal databases and support multi-user access with role-based permissions. The system aims to enhance efficiency, reduce delays, and improve the quality of legal services. However, it will not cover financial management or billing, as these are outside the scope of this project.

1.5 References

🎬 **ISO/IEC/IEEE 29148:2018** - Systems and software engineering — Life cycle processes — Requirements engineering
🎬 **IEEE Std 830-1998** - IEEE Recommended Practice for Software Requirements Specifications
🎬 **Pakistan Legal Framework** - Relevant legal texts and guidelines governing case management in Pakistan

2. Overall Description

2.1 Product Perspective

The **Legal Case Management System for Pakistan** is a state-of-the-art platform designed to revolutionize the management of legal cases, documents, and communication within legal institutions. This system represents a comprehensive solution tailored to meet the specific needs of law firms, courts, and government agencies, integrating modern technological advancements with user-centric design principles.

Integration and Compatibility:

The system is engineered to be a seamless extension of existing IT infrastructures within legal institutions. It is built to integrate effortlessly with:

- **Legacy Systems:** By using standardized APIs and data formats such as JSON and XML, the system ensures compatibility with current databases and software tools.
- **Third-Party Legal Software:** It supports integration with other legal tools and platforms, facilitating smooth data exchange and interoperability.
- **Existing Databases:** The system is capable of interfacing with various relational databases, allowing institutions to leverage their current data repositories.

System Architecture:

Adopting a **microservices architecture**, the system is divided into multiple independent modules, each responsible for specific functionalities such as:

- **Case Management**
- **Document Handling**
- **Scheduling**
- **Communication**

This architecture enhances scalability, allowing each component to be developed, updated, and maintained independently without affecting the overall system. It also supports fault isolation, ensuring that issues in one module do not compromise the entire system's functionality.

Deployment Models:

The platform offers flexible deployment options to cater to diverse institutional requirements:

1. **On-Premises:**
    - **Description:** Installed on local servers within the institution’s infrastructure.
    - **Advantages:** Full control over data and IT environment; customization of security measures to meet specific regulatory requirements.
    - **Considerations:** Requires physical hardware, maintenance, and in-house IT support.
2. **Cloud-Based:**
    - **Description:** Hosted on a secure, GDPR-compliant cloud platform.
    - **Advantages:** Scalability, reduced initial infrastructure costs, remote accessibility, and automatic updates.
    - **Considerations:** Dependent on internet connectivity; data security and compliance managed by cloud provider.
3. **Hybrid:**
    - **Description:** Combines on-premises and cloud solutions.
    - **Advantages:** Allows sensitive data to be stored locally while utilizing cloud resources for less critical functions or during peak times.
    - **Considerations:** Balances control and scalability; requires careful management of data flow between on-premises and cloud components.

2.2 Product Functions

The Legal Case Management System offers a suite of functionalities organized into distinct modules, each designed to address specific aspects of legal case management:

1. **Case Management Module:**
    - **Case Creation and Management:** Allows users to initiate new cases, assign case numbers, and input relevant details such as parties involved, case type, jurisdiction, and key dates.
    - **Case Status Tracking:** Provides real-time updates on case progress using visual indicators like timelines and progress bars.
    - **Historical Case Data:** Maintains a comprehensive audit trail of all case-related activities, including updates, document submissions, and communications.
2. **Document Management Module:**
    - **Document Storage:** Centralized, secure storage for various document formats (e.g., PDFs, Word documents, images).
    - **Version Control:** Tracks document changes, enabling users to view, revert to, or compare previous versions.
    - **Metadata and Tagging:** Facilitates categorization and retrieval through metadata and tagging.
    - **Access Control:** Implements role-based permissions to ensure that only authorized personnel can access, edit, or share sensitive documents.
3. **Court Scheduling Module:**
    - **Court Calendar:** Integrated calendar view displaying all upcoming court dates, hearings, and deadlines.
    - **Automated Notifications:** Sends reminders to legal professionals and clients about court dates, deadlines, and required actions.
    - **Conflict Management:** Identifies potential scheduling conflicts and suggests alternative dates or resource allocations.
4. **Client Communication Module:**
    - **Secure Messaging:** Enables encrypted communication between legal professionals and clients.
    - **Client Portal:** Provides clients with a dashboard to view case updates, submit documents, and communicate with their legal team.
    - **Appointment Scheduling:** Facilitates scheduling of meetings or consultations, integrated with the court scheduling module to avoid conflicts.
5. **Reporting and Analytics Module:**
    - **Customizable Reports:** Generates detailed reports on case progress, court schedules, client activity, and financial metrics.
    - **Data Analytics:** Provides insights into case trends, performance metrics, and resource utilization.

- **Export and Integration:** Supports exporting reports in various formats (e.g., PDF, Excel) and integrates with external analytics tools.

2.3 User Classes and Characteristics

The Case Management App features several user classes, each with specific roles and privileges tailored to different responsibilities within the system. The user classes are as follows:

- **Case Manager**
  - Responsibilities: Oversee case assignments, monitor case progress, manage client interactions, and coordinate with other professionals. They can create, edit, and close cases, assign tasks to other users, and generate reports.

- **Social Worker**
  - Responsibilities: Conduct client interviews, provide support and resources, and track client progress. They can access client case details, update client information, and log interactions and notes.
- **Client**
  - Responsibilities: View their case details, update personal information, and communicate with their case manager or social worker. They can submit requests for support or resources and track their progress through the case management system.
- **Administrator**
  - Responsibilities: Manage user accounts, configure system settings, and oversee overall system functionality. They can create and manage user accounts, assign roles, and generate system-wide reports.
- **Supervisor**
  - Responsibilities: Monitor the performance of case managers and social workers, review case progress, and ensure compliance with organizational policies. They can review and approve case changes and generate performance reports.
- **Program Coordinator**
  - Responsibilities: Manage specific programs or initiatives within the case management system. They can oversee program-related cases, track program performance, and coordinate with other users.
- **IT Support**
  - Responsibilities: Provide technical support to users, manage system maintenance, and handle troubleshooting. They can access system logs, perform system updates, and assist users with technical issues.

2.4 Operating Environment

The Case Management App will be accessible through both web and mobile platforms:

- **Web Application:**
  - Supported Browsers:
    - Google Chrome
    - Mozilla Firefox
    - Microsoft Edge
    - Safari
  - Developed using the MERN stack (MongoDB, Express.js, React.js, Node.js) for the web application.
- **Mobile Application:**
  - Platforms:
    - Android (version 7.0 or higher)
    - iOS (version 13 or higher)

   - Developed using Flutter and Dart for cross-platform compatibility.

The app will operate on a Cloud-Based Platform with the database managed through either MongoDB, Firebase, or a similar cloud-based solution. Integration with the server will use HTTPS, API endpoints, or VPN for secure data transactions.

2.5 Design and Implementation Constraints

The Case Management App development will encounter several constraints:

- **Legal and Ethical Constraints:** Compliance with data protection regulations (e.g., GDPR) is crucial. The system must ensure the confidentiality, integrity, and security of sensitive client information.
- **Security:** High priority will be given to implementing robust security measures, including data encryption, secure data transfer protocols, and comprehensive access controls to protect against unauthorized access.
- **Technical Constraints:** The app must handle concurrent user interactions and manage real-time data updates efficiently. This will require efficient coding practices and potentially complex multi-threading and asynchronous operations.
- **Maintenance and Updates:** The application will be version-controlled using GitHub. Regular updates and maintenance will be scheduled to ensure system reliability and security. Major updates will be applied during off-peak hours to minimize disruption.

2.6 User Documentation

User documentation will be provided to ensure smooth interaction with the Case Management App:

- **User Manual:** Available for all user levels, detailing how to use the app's features and functionalities.
- **Technical Documentation:** Includes detailed descriptions of system components, functionalities, and API usage for developers and technical staff.
- **Help Center:** An online help manual accessible via an "About" tab in both web and mobile applications. It will be updated with each new release and provide troubleshooting guides and FAQs.

2.7 Assumptions and Dependencies

Several assumptions and dependencies are made for the successful development of the Case Management App:

- **Technical Dependencies:** The app depends on modern web technologies (MERN stack) and cloud-based databases. Integration with these technologies will be seamless to ensure application performance.
- **User Assumptions:** It is assumed that end-users will have basic technical knowledge to operate the app. Training and support will be provided as needed.

- **Infrastructure:** The system will be networked and capable of routing through internet gateways. It is assumed that users will have access to the necessary hardware and internet connectivity.
- **Data Formats:** The app will handle various data formats, including images and documents. It is assumed that the infrastructure for managing these formats is in place.

3. External Interface Requirements

3.1 User Interfaces

The Case Management System will feature several user interfaces tailored to different user roles. These interfaces are designed to ensure an intuitive and efficient user experience:

- **Case Manager Interface:**
  - **Dashboard:** Provides an overview of assigned cases, upcoming tasks, and pending approvals.
  - **Case Management:** Tools for creating, editing, and closing cases. Includes case details, client information, and task assignment features.
  - **Reporting:** Generates reports on case progress, client interactions, and performance metrics.
- **Social Worker Interface:**
  - **Client Profiles:** Access and update client information, log interactions, and track client progress.
  - **Task Management:** Tools for scheduling and managing tasks related to client support and resource allocation.
  - **Notes and Records:** Interface for recording session notes and maintaining client records.
- **Client Interface:**
  - **Personal Dashboard:** View personal case details, update information, and track progress.
  - **Request Submission:** Submit requests for support or resources and communicate with case managers.
  - **Progress Tracking:** Access updates on case progress and view historical interactions.
- **Administrator Interface:**
  - **User Management:** Create and manage user accounts, assign roles, and configure system settings.
  - **System Reports:** Generate system-wide reports on user activity, system performance, and data integrity.
  - **Configuration:** Configure application settings and manage integration with external systems.
- **IT Support Interface:**

- **System Logs:** Access system logs and error reports for troubleshooting and maintenance.
- **Update Management:** Manage system updates and apply patches or fixes.
- **User Assistance:** Provide support to users and handle technical issues.

3.2 Hardware Interfaces

The Case Management System is a web application, so hardware interfaces are minimal and typically involve the following:

- **Server Hardware:**
  - **Web Servers:** Host the application and manage user requests.
  - **Database Servers:** Store and manage application data.
  - **Development Machines:** Used by developers for coding, testing, and deployment.
- **Client Hardware:**
  - **Desktops/Laptops:** Required for accessing the web application through browsers.
  - **Mobile Devices:** For clients and users accessing the system on mobile browsers.

3.3 Software Interfaces

The Case Management System will leverage a range of software interfaces to ensure seamless operation and data integration. The system will be built using the MERN stack, incorporating MongoDB for database management, Express.js for backend services, React.js for dynamic frontend development, and Node.js for server-side logic. For web scraping, tools like Firecrawl, Playwright, Puppeteer, and Scrapy-Splash will be employed to gather data from various sources efficiently. Automation will be managed through Make and GitHub Actions, streamlining build processes and continuous integration/deployment (CI/CD) workflows. Additionally, Apache Airflow will orchestrate data pipelines, handling Extract, Transform, Load (ETL) operations to ensure robust data management and integration.

3.4 Communications Interfaces

The Case Management System will utilize various communications interfaces to ensure secure and efficient data exchange. Secure data transfer will be managed through HTTPS protocols, protecting user data and ensuring encrypted communication between clients and servers. WebSockets may be employed for real-time updates, facilitating instant communication and notifications within the application. RESTful APIs will serve as the primary means for frontend-backend interaction, while GraphQL could be used for more flexible data querying. For external integrations, the system will interact with third-party APIs, and email notifications will be handled via SMTP to keep users informed about important updates and alerts.

4. System Features

4.1 Case Creation and Management

4.1.1 Create a New Case

4.1.1.1 Description and Priority

This feature allows users to create a new case in the system, capturing essential details for tracking and managing the case throughout its lifecycle.

4.1.1.2 Functional Requirements

R-CASE-1 The system shall allow users to enter case details including case title, description, priority level, and status.
R-CASE-2 The system shall allow users to assign a case to a specific team or individual.

R-CASE-3 The system shall allow users to set deadlines and milestones for the case.
R-CASE-4 The system shall store and display case creation date and time.

4.1.2 View Case Details

4.1.2.1 Description and Priority

This feature allows users to view detailed information about a specific case, including its history and current status.

4.1.2.2 Functional Requirements

R-CASE-5 The system shall display case details including title, description, priority, status, and assigned personnel.
R-CASE-6 The system shall display a history of updates and changes made to the case.
R-CASE-7 The system shall display deadlines and milestones associated with the case..

4.1.3 Update Case Information

4.1.3.1 Description and Priority

This feature allows users to update information related to a case, including its status, priority, and assigned team.

4.1.3.2 Stimulus/Response Sequences

The user selects a case to update.
The user modifies case details such as status, priority, or assignment.
The system saves the updated information and reflects these changes in the case history.

4.1.3.3 Functional Requirements

R-CASE-8 The system shall allow users to update case details including status, priority, and assigned personnel.
R-CASE-9 The system shall log all changes made to the case, including date, time, and user information

4.2 Case Assignment and Tracking

4.2.1 Assign Case

4.2.1.1 Description and Priority

This feature allows users to assign a case to specific team members or departments for further action.

4.2.1.2 Functional Requirements

R-ASSIGN-1 The system shall provide a list of available team members or departments for case assignment.
R-ASSIGN-2 The system shall allow users to assign or reassign cases to different individuals or teams.
R-ASSIGN-3 The system shall notify the assigned personnel of the new case assignment via email or in-app notification

4.2.2 Track Case Progres

4.2.2.1 Description and Priority

This feature allows users to track the progress of a case, including milestones achieved and pending tasks.

4.2.2.2 Functional Requirements

R-TRACK-1 The system shall provide a visual representation of case progress through milestones and tasks.
R-TRACK-2 The system shall allow users to update the progress of tasks and milestones.
R-TRACK-3 The system shall generate reports on case progress for review by managers or stakeholders.

4.2.3 Case Resolution and Closure

4.2.3.1 Description and Priority

This feature allows users to mark a case as resolved once all tasks and objectives have been completed.

4.2.3.2 Functional Requirements

R-RESOL-1 The system shall allow users to mark a case as resolved.
R-RESOL-2 The system shall require users to provide a resolution summary or closure notes.
R-RESOL-3 The system shall update the case status to "Resolved" and lock further modifications.

4.3 Case Resolution and Closure

4.3.1 Resolve Case

4.3.1.1 Description and Priority

This feature allows users to mark a case as resolved once all tasks and objectives have been completed.

4.3.1.2 Functional Requirements

R-RESOL-1 The system shall allow users to mark a case as resolved.
R-RESOL-2 The system shall require users to provide a resolution summary or closure notes.
R-RESOL-3 The system shall update the case status to "Resolved" and lock further modifications.

4.3.2 Close Case

4.3.2.1 Description and Priority

This feature allows users to officially close a case, finalizing all related documentation and tasks.

4.3.2.2 Functional Requirements

R-CLOSE-1 The system shall allow users to close a case once it is marked as resolved.
R-CLOSE-2 The system shall archive case details and related documents upon closure.
R-CLOSE-3 The system shall generate a final report summarizing the case resolution and actions taken.

4.3.3 Case Audit and Review

4.3.3.1 Description and Priority

This feature allows users to review and audit cases post-closure to ensure compliance and effectiveness of resolution.

4.3.3.2 Functional Requirements

R-AUDIT-1 The system shall allow users to review case history and final resolution details.
R-AUDIT-2 The system shall provide tools for generating audit reports.
R-AUDIT-3 The system shall allow users to review case performance metrics and resolution effectiveness.

5. User Stories

5.1 Admin User Stories

5.1.1 User Management

As an Admin,
I want to be able to manage user accounts (e.g., add, remove, update, deactivate users)
So that I can control access and roles within the system.

- **Preconditions:** Admin is logged in.
- **Main Flow:**
    1. Admin selects the user management option.
    2. Admin adds new users (e.g., judges, lawyers, researchers).
    3. Admin updates user roles and permissions.
    4. Admin deactivates or deletes users.
    5. Changes are saved in the system.

5.1.2 Generate Case Reports

As an Admin,
I want to generate reports on cases managed in the system
So that I can analyze case trends and performance.

- **Preconditions:** Admin is logged in, and cases exist in the system.
- **Main Flow:**
  1. Admin selects the report generation feature.
  2. Admin selects filters (e.g., time range, case type, case status).
  3. System generates the report.
  4. Admin can view or export the report as a PDF or CSV.

5.1.3 Audit Trail Management

As an Admin,
I want to access audit logs of actions taken by different users
So that I can track and review user activities for accountability.

- **Preconditions:** Admin is logged in.
- **Main Flow:**
  1. Admin selects the audit trail option.
  2. Admin reviews actions taken by users (e.g., case updates, note additions).
  3. Admin filters logs by user, date, or type of action.
  4. Logs are saved or archived for future reference.

5.1.4 System Health Monitoring

As an Admin,
I want to monitor the system's health (e.g., database status, server uptime, security breaches)
So that I can maintain system performance and handle issues promptly.

- **Preconditions:** Admin is logged in.
- **Main Flow:**
  1. Admin selects the system health option.
  2. Admin checks the status of servers, storage usage, and potential breaches.
  3. Admin resolves alerts or escalates issues to IT teams.

5.2 Lawyer User Stories

5.2.1 File Legal Motions

As a Lawyer,
I want to file legal motions related to cases
So that I can ensure the case progresses through the court system.

- **Preconditions:** Lawyer is logged in and working on an active case.
- **Main Flow:**
    1. Lawyer selects the case and navigates to the motions section.
    2. Lawyer fills in the details for the motion.
    3. System logs and submits the motion to the case record.
    4. Notifications are sent to relevant parties (e.g., judge, opposing lawyer).

5.2.2 Schedule Hearings

As a Lawyer,
I want to schedule hearings related to my cases
So that I can manage court proceedings efficiently.

- **Preconditions:** Lawyer is logged in and has access to cases.
- **Main Flow:**
    1. Lawyer selects the case for which they want to schedule a hearing.
    2. Lawyer inputs the hearing details, including date and participants.
    3. System confirms availability and schedules the hearing.
    4. Hearing is added to the case log, and all participants are notified.

5.3 Judge User Stories

5.3.1 Assign Verdicts

As a Judge,
I want to assign a verdict for a case after hearings are concluded
So that I can officially close the case.

- **Preconditions:** Judge is logged in, and hearings for the case are completed.
- **Main Flow:**
    1. Judge navigates to the case's verdict section.
    2. Judge enters the final verdict details (e.g., guilty, not guilty, penalties).
    3. System updates the case status to "Closed" and records the verdict.
    4. All parties involved in the case are notified of the decision.

5.3.2 Review Case History

As a Judge,
I want to review the complete history of a case before making a decision
So that I can ensure informed judgment.

- **Preconditions:** Judge is logged in and has access to case details.
- **Main Flow:**
    1. Judge selects a case to review.
    2. Judge navigates through all previous hearings, notes, and legal documents.
    3. Judge searches specific logs or documents.
    4. Judge uses this information for case decisions.

5.4 Legal Professional User Stories

5.4.1 Submit Evidence

As a Legal Professional,
I want to submit evidence to a case record
So that I can ensure all necessary information is documented.

- **Preconditions:** Legal professional is logged in and has access to the case.
- **Main Flow:**

1. Legal professional navigates to the evidence section.
2. Legal professional uploads documents, images, or videos as evidence.
3. System logs the evidence and associates it with the case.
4. Notification is sent to the case participants about the new evidence.

5.5 Compliance Bot User Stories

5.5.1 Ask Legal Compliance Questions

As a User (Admin, Lawyer, Legal Researcher, Judge, Legal Professional),
I want to ask the compliance bot legal questions related to laws and regulations
So that I can receive legal references quickly.

- **Preconditions:** User is logged in and has access to the compliance bot.
- **Main Flow:**
    1. User enters a legal question into the bot.
    2. Compliance bot searches legal databases and references.
    3. Bot returns relevant legal statutes, regulations, and case law.
    4. User reviews the information for case use.

5.6 Assistance Bot User Stories

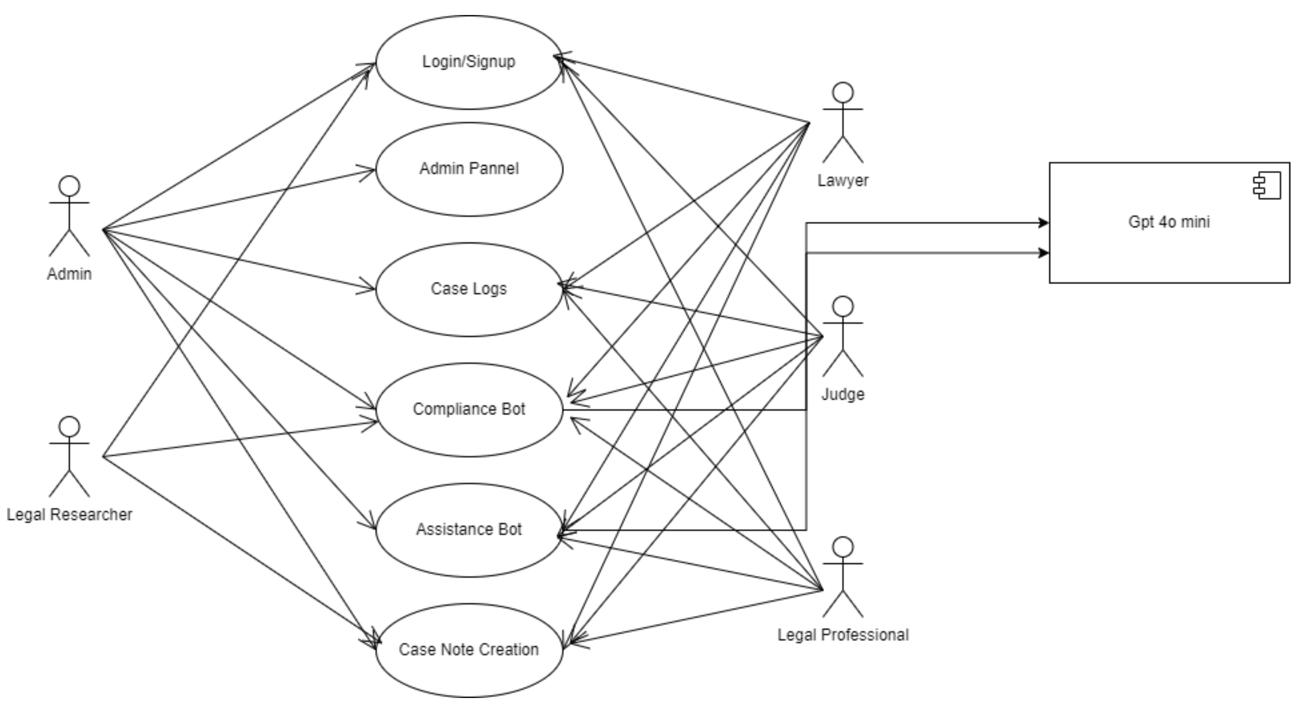
5.6.1 Ask Case-Specific Questions

As a User (Admin, Lawyer, Judge, Legal Professional),
I want to ask the assistance bot questions about specific cases
So that I can quickly retrieve relevant case information.

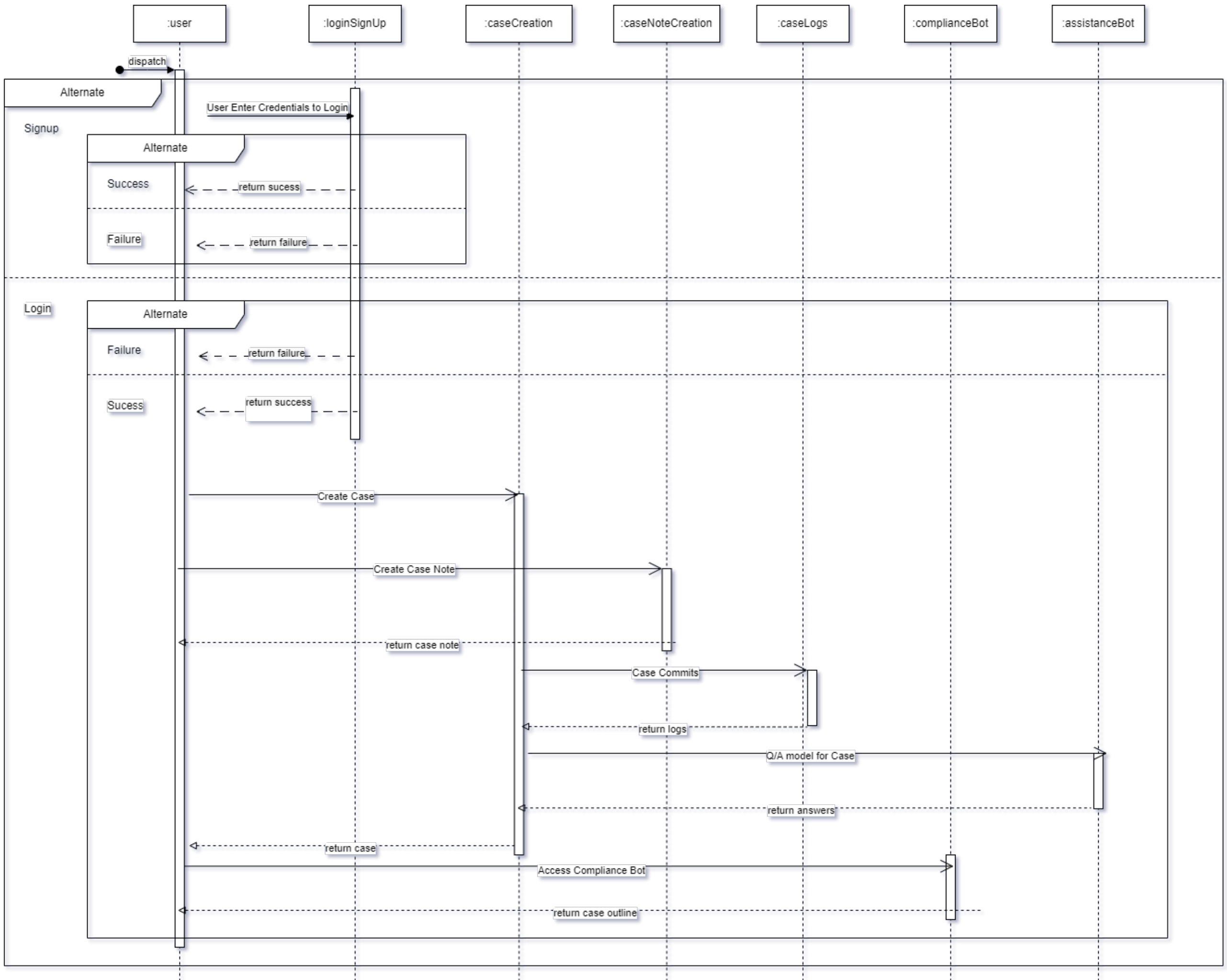
- **Preconditions:** User is logged in and has access to the case.
- **Main Flow:**
    1. User asks the bot for a case summary or specific details (e.g., hearing dates, parties involved).
    2. Assistance bot retrieves the requested case data from the system.
    3. User receives relevant information and insights.

6. Diagrams

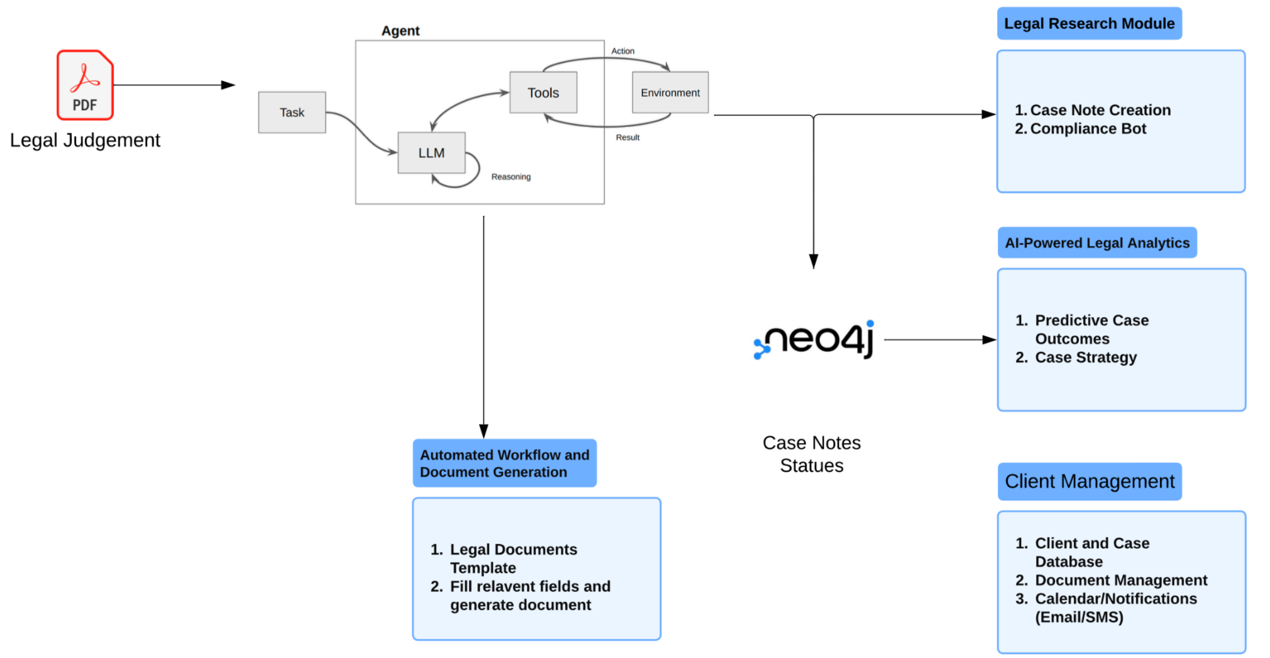
6.1 Use Case Diagram



6.2 Sequence Diagram



Project Architecture



7. Other Nonfunctional Requirements

7.1 Performance Requirements

The system must exhibit robust performance characteristics to ensure a seamless user experience. Specifically, the response time for user interactions, including case creation, updates, and retrieval, should be optimized to complete within 3 seconds under normal operating conditions. This swift response time is essential to maintain operational efficiency and user satisfaction. Furthermore, the system must be capable of processing and displaying search results for up to 10,000 cases within 5 seconds, ensuring that users can quickly access the information they need.

Scalability is another critical aspect of performance. The system should be designed to accommodate up to 100,000 active cases without noticeable performance degradation. It must also support up to 1,000 concurrent users while maintaining smooth and responsive interactions. To manage high volumes of case activity, the system should handle at least 200 case updates per minute and support up to 500 new case entries per hour.

Resource utilization must be managed effectively to avoid performance bottlenecks. The system should ensure that CPU usage does not exceed 70% during peak load conditions, and memory usage should be optimized to prevent the average memory footprint from surpassing 2 GB per active user session.

7.2 Safety Requirements

Ensuring the safety of the system involves implementing robust data integrity and error-handling measures. The system must incorporate mechanisms to guarantee data integrity, such as transaction rollback and recovery procedures in the event of system failures. This ensures that case records remain accurate and reliable. Additionally, data validation checks should be employed to prevent the entry of invalid or corrupted data.

Error handling is crucial for maintaining system stability. The system should provide detailed error messages and logging to facilitate troubleshooting and resolution of issues. Automatic error recovery procedures should be implemented to minimize the impact of unexpected failures or disruptions, thus maintaining system reliability and continuity.

The system should also be designed with a high level of reliability, maintaining an uptime of 99.9% to ensure continuous availability. Regular backup procedures must be in place to protect case data, allowing for restoration in the event of hardware or software failures.

7.3 Security Requirements

Security is a paramount concern for the system, necessitating comprehensive measures to protect sensitive information. User authentication must be secured with a combination of usernames and passwords, and the system should support multi-factor authentication (MFA) to enhance security. Role-based access control (RBAC) must be implemented to restrict access to system features based on user roles and permissions, ensuring that only authorized personnel can perform specific actions.

Data encryption is essential for safeguarding sensitive information. The system must utilize industry-standard encryption algorithms, such as AES-256, to protect data both in transit and at rest. Secure communication protocols, like HTTPS, should be employed to protect data transmitted between the client and server.

Audit and logging mechanisms are crucial for monitoring and securing the system. The system should maintain an audit trail of user actions and system events, such as case creation, updates, and deletions, to support compliance and forensic investigations. Security-related events, including failed login attempts and unauthorized access attempts, should also be logged.

Regular security assessments and vulnerability scans are necessary to identify and address potential security risks. The system should include mechanisms for applying security patches and updates promptly to protect against known vulnerabilities.

7.4 Software Quality Attributes

The quality attributes of the system encompass usability, maintainability, extensibility, and performance. Usability is a key focus, and the system should offer an intuitive and user-friendly interface to facilitate ease of use, even for users with limited technical expertise. Context-sensitive help and user documentation must be provided to assist users in navigating and utilizing system features effectively.

Maintainability is another important attribute. The system should be designed with a modular architecture to support easy maintenance and updates. Comprehensive documentation, including code comments and design specifications, should be available to aid in ongoing maintenance and troubleshooting efforts.

Extensibility is crucial for accommodating future enhancements and additional features. The system should be designed to support future integrations and new functionalities without requiring major modifications. APIs and integration points should be included to facilitate seamless connections with other systems and applications.

Performance quality attributes must be addressed to ensure efficient operation. The system should be optimized for performance, ensuring that all operations, including case management and reporting, are executed efficiently within acceptable time limits. Performance monitoring tools should be incorporated to track and analyze system performance, helping to identify areas for improvement.

7.5 Business Rules

The system must adhere to several business rules to ensure effective case management. Case priority levels are a fundamental aspect, with support for multiple levels including Low, Medium, High, and Urgent. This functionality facilitates the effective management of cases and resource allocation based on urgency and importance. Users should be able to set and modify priority levels as needed.

Case assignment rules are essential for routing cases to appropriate personnel or teams based on their expertise and availability. The system should enforce these assignment rules and include automatic assignment features to streamline the process and ensure cases are directed to the most suitable individuals.

Resolution criteria for cases must be clearly defined. The system should ensure that all required tasks are completed, resolution details are documented, and approval from relevant stakeholders is obtained before allowing a case to be closed. Validation checks should be in place to confirm that all resolution criteria are met.

Reporting and compliance are also critical. The system should generate reports on case status, progress, and performance to support compliance with organizational policies and regulatory requirements. Features for customizing reports and exporting data should be included to address various business and compliance needs.

8. Other Requirements

In addition to the core functional and nonfunctional requirements, several other requirements must be addressed to ensure the effective operation and integration of the Case Management System.

These requirements include compliance with legal and regulatory standards, user training, and system documentation.

Compliance Requirements: The system must comply with relevant legal and regulatory standards applicable to case management and data protection. This includes adherence to data protection regulations such as the General Data Protection Regulation (GDPR) or local data protection laws, ensuring that personal and sensitive information is handled in accordance with legal requirements. The system should also comply with industry-specific regulations, such as healthcare compliance standards for handling medical information.

User Training: To ensure that users can effectively utilize the system, comprehensive training programs must be developed and delivered. These training programs should cover all aspects of the system, including navigation, case management procedures, and reporting functionalities. Training materials should be provided in various formats, such as manuals, online tutorials, and hands-on workshops. Additionally, a helpdesk or support team should be available to assist users with any issues or questions that arise during and after the training period.

System Documentation: Detailed system documentation is essential for both users and developers. For users, documentation should include user guides, FAQs, and troubleshooting tips to facilitate smooth operation. For developers and system administrators, technical documentation should provide an in-depth overview of the system architecture, design decisions, and codebase. This documentation should also include maintenance procedures, integration points, and any customizations made to the system. Regular updates to the documentation should be made to reflect changes and improvements to the system.

Integration Requirements: The system must be capable of integrating with other existing systems and applications within the organization. This includes compatibility with data sources, such as databases and third-party applications, and support for data exchange formats and protocols. Integration should be seamless to ensure that case data can be shared and utilized effectively across different platforms.

Accessibility Requirements: The system should be designed with accessibility in mind to ensure that users with disabilities can interact with it effectively. This includes compliance with accessibility standards, such as the Web Content Accessibility Guidelines (WCAG), and incorporating features such as keyboard navigation, screen reader compatibility, and high-contrast modes.

Backup and Recovery Requirements: To safeguard against data loss and ensure business continuity, the system must include robust backup and recovery mechanisms. Regular backups of case data should be performed and stored securely. In the event of data loss or system failure, recovery procedures should be in place to restore data and resume operations with minimal disruption

Appendix A: Glossary

1. **Case**: An individual matter or issue being tracked and managed within the system. It includes all related details, documents, and interactions.
2. **Case Management System (CMS)**: A software platform designed to streamline the process of managing legal cases, including tracking, documentation, scheduling, and communication.
3. **Client Portal**: A web-based interface through which clients can view their case details, update personal information, and communicate with their legal team.
4. **Microservices Architecture**: An architectural style that structures an application as a collection of loosely coupled services, each of which implements a specific business capability.
5. **API (Application Programming Interface)**: A set of rules and tools for building software and applications, allowing different systems to communicate with each other.
6. **GDPR (General Data Protection Regulation)**: A regulation in EU law on data protection and privacy for all individuals within the European Union and the European Economic Area.
7. **RESTful API**: An architectural style for designing networked applications, relying on HTTP requests to access and use data.
8. **GraphQL**: A query language for APIs and a server-side runtime for executing those queries with existing data.
9. **WebSocket**: A protocol for full-duplex communication channels over a single TCP connection, commonly used in real-time web applications.
10. **HTTPS (HyperText Transfer Protocol Secure)**: An extension of HTTP that uses SSL/TLS to provide secure communication over a computer network.

Appendix B: Acronyms

1. **API**: Application Programming Interface
2. **CMS**: Case Management System
3. **GDPR**: General Data Protection Regulation
4. **HTTP**: HyperText Transfer Protocol
5. **HTTPS**: HyperText Transfer Protocol Secure
6. **REST**: Representational State Transfer
7. **SSL**: Secure Sockets Layer
8. **TLS**: Transport Layer Security
9. **UI**: User Interface
10. ***UX****: User Experience*

Appendix C: To Be Determined List

Business Rules
Other Requirements
Currently the added data for above isn't finalized

Appendix D: Change Log

Date	Version	Description	Author
2024-10-4	1.0	Initial release of the Software Requirements Specification	Muhammad Mahad Sheikh, Wajeeh Ul Hassan